D108-D109 | Lower cost and winning can mislead

PRIVATE SYNTHETIC annual cement configuration and separate same-date spectrum-style common-value auction. No actual India market, rule, company, order or bid.

Attempt before reference	Derive annual output/cost/threshold comparisons and separately the joint-state win-conditioned values. Attempt without notes first. Actual confidence, errors, reviewer words and delayed transfer remain blank.
D108 | Output is not guaranteed	Current capacity 100000/output 80000/price 5000/variable 3000/fixed 120 million -> result 40 million. Expanded capacity 150000/variable 2800/fixed 180 million gives minus 4 at sales 80000, or 84 at hypothetical sales 120000. Annual operating result, not project NPV.
D108 | Match the feasible domain	Same-output cost difference is 60 million minus 200 Q; equality 300000 is outside common capacity 100000. Expanded matches old result at 100000 tonnes; zero-result needs 81818.181818 expected or 81819 whole tonnes. Lower marginal cost need not lower average cost.
D109 | Condition on the win event	Low/high value 80/160 with equal priors; rival 90/140 with probabilities conditional on low .75/.25, high .25/.75. Bids 105/120 both win on rival 90 only. Pwin .5/posterior high .25/conditional common 100, not prior 120.
D109 | Candidates and risk	Net private benefit 20-10=10 -> conditional net 110. Bid 105 gives conditional 5/expected 2.5/low-state loss 15; bid 120 gives minus 10/minus 5. Required conditional surplus 5 and worst loss 20 bound bid 105 in the strict 90-140 interval. Best supplied candidate is not global optimum.
Limits and scope	No actual plant expansion, auction application, deposit, financing, competitor contact, coordination, disclosure or spending authorized. Static prose is Case-v 1 and must change after inputs. Native arithmetic is not source/AV, actual learner, cross-viewer, legal or overall 10 clearance.

D108 inputs | Annual operating cases

Tonnes/year, rupees/tonne and MILLION rupees total. Annual fixed costs are not initial capex or automatically sunk. Other investment cash flows are unknown.

Current annual capacity, tonnes	100000.00
Current annual output/sales, tonnes	80000.00
Common base price, rupees/tonne	5000.00
Current variable cost, rupees/tonne	3000.00
Current annual fixed operating cost, million rupees	120.00
Expanded annual capacity, tonnes	150000.00
Expanded annual fixed operating cost, million rupees	180.00
Expanded variable cost, rupees/tonne	2800.00
Separate expanded demand-case sales, tonnes	120000.00
Separate response-case price, rupees/tonne	4600.00

Plant structural inputs valid? (1=yes)	1.000000
Current output feasible? (1=yes)	1.000000

Structural validity only, not real demand, rights, cost evidence, authority or investment feasibility. Negative computed results stay visible. Yellow cells are synthetic assumptions, not current observations.

D108 cases | Separate outputs and prices

Annual million rupees except unit price/cost and tonnes. Fixed cost included once in each case. No initial capex, tax, working capital, finance, lifetime or residual value, so not NPV.

Measure	Current	Expanded same sales	Expanded demand case	Response price case
Capacity, tonnes/year	100000.000000	150000.000000	150000.000000	150000.000000
Output/sales, tonnes/year	80000.000000	80000.000000	120000.000000	120000.000000
Price, rupees/tonne	5000.000000	5000.000000	5000.000000	4600.000000
Variable cost, rupees/tonne	3000.000000	2800.000000	2800.000000	2800.000000
Fixed operating cost, million	120.000000	180.000000	180.000000	180.000000
Output within capacity? (1=yes)	1.000000	1.000000	1.000000	1.000000
Utilization	80.0000%	53.3333%	80.0000%	80.0000%
Revenue, million	400.000000	400.000000	600.000000	552.000000
Variable expense, million	240.000000	224.000000	336.000000	336.000000
Total cost, million	360.000000	404.000000	516.000000	516.000000
Average cost, rupees/tonne	4500.000000	5050.000000	4300.000000	4300.000000
Operating result, million	40.000000	-4.000000	84.000000	36.000000
Result change vs current, million	0.000000	-44.000000	44.000000	-4.000000
Marginal production cost, rupees/tonne	3000.000000	2800.000000	2800.000000	2800.000000

Current 40; expanded same sales -4; demand case 84; response 36. Average 4500 at 80000 versus 4300 at 120000 is not a same-output saving. Capacity is a constraint, not proof of demand. MC is constant only in this stipulated range, not the discrete cost of adding a plant.

D108 thresholds | Required is not available

Million rupees converted to rupees before dividing by unit margin. Thresholds require strictly positive unit margin; zero-output average cost undefined. Output requirements do not supply demand or approvals.

Measure	Base price	Response price
Expanded unit contribution, rupees/tonne	2200.000000	1800.000000
Expected tonnes for zero operating result	81818.181818	100000.000000
Minimum whole tonnes for zero result	81819.000000	100000.000000
Expected tonnes to match CURRENT result	100000.000000	122222.222222
Minimum whole tonnes to match current	100000.000000	122223.000000
Required utilization to match current	66.6667%	81.4815%
Match-current threshold within capacity? (1=yes)	1.000000	1.000000

Common feasible output upper bound, tonnes	100000.000000
Same-output cost equality tonnes, if slopes differ	300000.000000
Equality inside common range? (1=yes)	0.000000
Expanded-minus-current cost at Q=0, million	60.000000
Expanded-minus-current cost at common max, million	40.000000
Expanded no more costly across common range? (1=yes)	0.000000

Baseline cost equality 300000 is infeasible; difference endpoints 60/40 million show expanded costs more throughout common output 0-100000. If slopes equal, equality quantity undefined; endpoints still classify cost ordering. Annual fixed operating cost is not capex or long-run optimized scale evidence.

D109 inputs | A joint distribution, not a haircut

One date, crore. Winner pays own bid and must bid STRICTLY above rival maximum, ties lose. No losing costs. Candidate comparison is not global strategy optimization. Net private benefit applies only if won.

Low common value, crore present value	80.00
High common value, crore present value	160.00
Prior high probability	0.50
Lower rival maximum bid, crore	90.00
Higher rival maximum bid, crore	140.00
P(lower rival bid | low value)	0.75
P(lower rival bid | high value)	0.25
Private operating synergy, crore PV	20.00
Integration cost, crore PV	10.00
Supplied candidate bid 1, crore	105.00
Supplied candidate bid 2, crore	120.00
Required expected surplus CONDITIONAL on win	5.00
Maximum realized winning loss	20.00

Auction structural inputs valid? (1=yes)	1.000000

Structural validity only, not real demand, rights, cost evidence, authority or investment feasibility. Negative computed results stay visible. Yellow cells are synthetic assumptions, not current observations.

D109 joint states | Win selection changes value

Four exhaustive states. State probabilities, not independent marginal products. Candidate win flags use strict bid>rival, including changed bids outside the original interval.

State	Joint probability	Common value PV	Rival maximum bid	Net own value PV	Bid 1 wins?	Bid 2 wins?
Low / lower rival	37.5000%	80.000000	90.000000	90.000000	1.000000	1.000000
Low / higher rival	12.5000%	80.000000	140.000000	90.000000	0.000000	0.000000
High / lower rival	12.5000%	160.000000	90.000000	170.000000	1.000000	1.000000
High / higher rival	37.5000%	160.000000	140.000000	170.000000	0.000000	0.000000
Probability sum	100.0000%					

Baseline joint probabilities .375/.125/.125/.375. Both supplied bids win only rows 6/8: probability .5. High-winning mass .125, posterior .25, not prior .5. Changed bids must recompute win events; tie rule is stipulated, not a verified auction regulation.

D109 candidates | Expected gain and downside differ

All payoff/value amounts crore PV. Abstain is explicit zero. No-win conditional quantities are undefined, not zero. Tests classify supplied candidates only, not an optimal bidding strategy.

Measure	Supplied bid 1	Supplied bid 2
Bid price	105.000000	120.000000
Win probability	50.0000%	50.0000%
High AND win probability	12.5000%	12.5000%
P(high | win)	25.0000%	25.0000%
Common value conditional on win	100.000000	100.000000
Net own value conditional on win	110.000000	110.000000
Expected surplus conditional on win	5.000000	-10.000000
Unconditional expected surplus	2.500000	-5.000000
Worst realized winning loss, nonnegative	15.000000	30.000000
Conditional surplus minimum met? (1=yes)	1.000000	0.000000
Worst loss limit met? (1=yes)	1.000000	0.000000
Both constraints met? (1=yes)	1.000000	0.000000
Abstain expected surplus	0.000000	0.000000
Highest supplied expected surplus incl abstain	2.500000	2.500000

Baseline bid 105 expected 2.5 versus bid 120 minus 5, abstain 0; low-winning loss 15 versus 30. Engineering correction synergy 10 yields bid 105 expected minus 2.5, so abstain beats both supplied bids. Neither expected advantage nor a risk flag supplies bid authority or removes realized losses.

D109 bounds | Recompute outside the interval

Bounds apply ONLY to the strict lower<bid<higher rival interval with positive winning probability. They are conditional decision constraints, not a new supplied candidate, global optimum or actual instruction.

Win probability in strict middle interval	50.0000%
Posterior high in interval	25.0000%
Interval conditional common value	100.000000
Interval conditional net own value	110.000000
Expected conditional break-even bid bound	110.000000
Bid bound for required conditional surplus	105.000000
Worst-loss bid bound over positive-prob win states	110.000000
Combined conditional upper bound	105.000000
At least one strict-interval bid can meet bound?	1.000000

Baseline expected bound 110, surplus bound 105, worst-loss bound 110 -> combined 105. Must ALSO stay 90<bid<140. If combined bound>=140 it does not authorize bidding outside interval; strict upper endpoint remains 140. If both winning masses zero, conditional bounds undefined. With synergy 10, combined 95, not a new approved candidate.

Audit | Arithmetic identity is not authority

Separate models and units. A broken result or probability reference blocks affected summary outputs, not the independent case. Neither a flag nor positive expected payoff authorizes investment or bidding.

Plant cost/result identities valid? (1=yes)	1.000000
Guarded current operating result, million	40.000000
Joint-state probability/value identity valid? (1=yes)	1.000000
Guarded bid 1 expected surplus, crore	2.500000

Real capex, demand, financing, rights, legal rules, rival distribution and valuation evidence remain absent. The interval bound cannot be carried across a changed win event. Static prose needs revision after input edits. Actual learner or independent human review was not performed.

Evidence | Preserve the unknowns

Original D108/D109 synthetic exercises at https://top1os.dev/. Existing selected OpenStax/Yale teaching retained in the course; no new empirical reading or full audiovisual review claimed.

D108 | Units and dependencies	Plant inputs -> Plant cases/Thresholds -> Audit. One year; annual output tonnes, price/cost rupees per tonne, total money million rupees. Capex, tax, working capital, lifetime and residual absent. Annual fixed operating cost is not the investment amount.
D109 | Units and dependencies	Auction inputs -> Joint states/Candidates/Interval bounds -> Audit. One present-value date, crore. Strict own-bid>rival; ties lose; no losing cost. Joint probabilities encode correlation, not independent rival estimates. No actual auction regulation verified.
Strongest authored objections	D108 volume may not exist; cost absorption is not an optimized long-run scale curve. D109 rival information/distribution and private synergy may be wrong; expected gain permits losses. Preserve contradictory evidence and recompute only affected inputs, not a generic haircut.
Fictional day 0/7/30 review	Engineering/demand/cash roles check capacity, annual cost and supported demand before hypothetical investment recommendation. Engineering/valuation/auction roles check joint state and net private benefit. No real owners accepted these roles or supplied evidence/permission.
Stop gates	Stop affected real work without actual demand/capex/timing/rights/rules and authority. No real plant order, auction application, deposit, finance, customer/competitor contact, collusion or disclosure. A synthetic result, mock sponsor or bid ceiling creates no authority.
Changed-context delayed transfer	Without notes lower response price and separately alter joint high prior, bid interval and synergy. Rebuild feasibility, denominators and win selection. Record actual reasoning/error/source/correction/reviewer objection and delayed novel result only if performed.

Actual work | No performed learner or review evidence

Record actual attempt, confidence, error, correction, strongest actual objection and delayed new context only if performed. No real reviewer/learner result is supplied.

Actual date / case	Attempt / confidence	Exact source / input	Observed error / claim	Correction / changed decision	Actual objection / response	Delayed context / result

Actual work remains blank. Record only what occurred: dates, reasoning, error/source/correction, strongest objection and a new delayed transfer. No invented learner results, sponsor acceptance, actual budget release or signed rights.

Failure log | Authored execution only

Observed native tests are added only after execution. Not actual learner mistakes, independent human audit, Excel/Sheets parity, real authority or overall 10/10.

Controlled test	Observed authored outcome
Baseline plant / auction identities	PASS: all baseline formulas independently recalculated, annual and PV cases separate.
Response price 4400 / restore	PASS: response 4400 changes result/required output, baseline restored.
Sales 150000 and infeasible 150001	PASS: sales 150000 capacity equality valid;150001 blocks infeasible outputs.
Equal costs / zero unit margin	PASS: equal costs leave cost-equality quantity undefined; zero margin blocks thresholds.
Synergy 10 / restore	PASS: synergy 10 makes supplied bids lose expected value; abstain dominates.
Bid 150 all-win / bid 90 no-win	PASS: bid 150 recomputes all-win posterior;bid 90 no-win conditional outputs undefined.
Changed prior / rival correlation	PASS: changed prior/correlation and high-only/no-middle-win states recomputed.
Invalid inputs / case isolation	PASS: 12 invalid cases fail closed while independent model preserved.
Broken plant / joint refs	PASS: broken plant result and joint probability block affected audit summaries.
Restored baseline / actual blanks	PASS: baseline restored, saved caches error-free, actual learner/reviewer/delayed blank.

Only the original synthetic course exercise at https://top1os.dev/ supplies the case figures. No new empirical reading, real participants, company records, Indian legal clearance, actual commitments or source completeness claimed.